

Chapter

6



Cost-Volume-Profit Relationships

Cost-Volume-Profit (CVP)

CVP analysis is a powerful tool that helps managers understand the relationships among cost, volume and profit. CVP analysis focuses on how profits are affected by the following five factors:

- Selling prices
- Sales volume
- Unit variable costs
- Total fixed costs
- Mix of products sold

The Basics of Cost-Volume-Profit (CVP) Analysis

WIND BICYCLE CO.		
Contribution Income Statement		
For the Month of June		
	Total	Per Unit
Sales (500 bikes)	\$250,000	\$ 500
Less: variable expenses	150,000	300
Contribution margin	100,000	\$ 200

Contribution Margin (CM) is the amount remaining from sales revenue after variable expenses have been deducted.



The Basics of Cost-Volume-Profit (CVP) Analysis

WIND BICYCLE CO. Contribution Income Statement For the Month of June		
	Total	Per Unit
Sales (500 bikes)	\$250,000	\$ 500
Less: variable expenses	150,000	300
Contribution margin	100,000	\$ 200
Less: fixed expenses	80,000	
CM goes to cover fixed expenses.		



The Basics of Cost-Volume-Profit (CVP) Analysis

WIND BICYCLE CO.		
Contribution Income Statement		
For the Month of June		
	Total	Per Unit
Sales (500 bikes)	\$250,000	\$ 500
Less: variable expenses	150,000	300
Contribution margin	100,000	\$ 200
Less: fixed expenses	80,000	
Net income	\$ 20,000	

After covering fixed costs, any remaining CM contributes to income.



The Contribution Approach

Consider the following information developed by the accountant at Wind Bicycle Co.:

	<u>Total</u>	<u>Per Unit</u>
Sales (500 bikes)	\$250,000	\$ 500
Less: variable expenses	150,000	300
Contribution margin	\$100,000	<u>\$ 200</u>
Less: fixed expenses	80,000	
Net income	<u>\$ 20,000</u>	



The Contribution Approach

For each additional unit Wind sells, **\$200** more in contribution margin will help to cover fixed expenses and profit.

	Total	Per Unit
Sales (500 bikes)	\$250,000	\$ 500
Less: variable expenses	150,000	300
Contribution margin	\$100,000	\$ 200
Less: fixed expenses	80,000	
Net income	\$ 20,000	



The Contribution Approach

Each month Wind must generate at least **\$80,000** in total CM to break even.

	<u>Total</u>	<u>Per Unit</u>
Sales (500 bikes)	\$250,000	\$ 500
Less: variable expenses	<u>150,000</u>	<u>300</u>
Contribution margin	\$100,000	<u>\$ 200</u>
Less: fixed expenses	<u>80,000</u>	
Net income	<u><u>\$ 20,000</u></u>	



The Contribution Approach

If Wind sells 400 units in a month, it will be operating at the break-even point.

WIND BICYCLE CO. Contribution Income Statement For the Month of June		
	Total	Per Unit
Sales (400 bikes)	\$200,000	\$ 500
Less: variable expenses	120,000	300
Contribution margin	80,000	<u>\$ 200</u>
Less: fixed expenses	80,000	
Net income	<u><u>\$ 0</u></u>	



The Contribution Approach

If Wind sells one additional unit (401 bikes), net income will increase by **\$200.**

WIND BICYCLE CO.		
Contribution Income Statement		
For the Month of June		
	Total	Per Unit
Sales (401 bikes)	\$200,500	\$ 500
Less: variable expenses	120,300	300
Contribution margin	80,200	\$ 200
Less: fixed expenses	80,000	
Net income	\$ 200	



The Contribution Approach

The break-even point can be defined either as:

- The point where total sales revenue equals total expenses (variable and fixed).
- The point where total contribution margin equals total fixed expenses.

Contribution Margin Ratio

The contribution margin **ratio** is:

$$\text{CM Ratio} = \frac{\text{Contribution margin}}{\text{Sales}}$$

For Wind Bicycle Co. the ratio is:

$$\frac{\$200}{\$500} = 40\%$$

Contribution Margin Ratio

At Wind, each \$1.00 increase in sales revenue results in a total contribution margin increase of 40¢.

If sales increase by \$50,000, what will be the increase in total contribution margin?

Contribution Margin Ratio

	400 Bikes	500 Bikes
Sales	\$200,000	\$250,000
Less: variable expenses	120,000	150,000
Contribution margin	80,000	100,000
Less: fixed expenses	80,000	80,000
Net income	\$ -	\$ 20,000

A \$50,000 increase in sales revenue

Contribution Margin Ratio

	<u>400 Bikes</u>	<u>500 Bikes</u>
Sales	\$200,000	\$250,000
Less: variable expenses	120,000	150,000
Contribution margin	80,000	100,000
Less: fixed expenses	80,000	80,000
Net income	\$ -	\$ 20,000

**A \$50,000 increase in sales revenue
results in a \$20,000 increase in CM.
(\$50,000 × 40% = \$20,000)**

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. 2,100 cups are sold each month on average. What is the CM Ratio for Coffee Klatch?

- a. 1.319
- b. 0.758
- c. 0.242
- d. 4.139

Changes in Fixed Costs and Sales Volume

Wind is currently selling 500 bikes per month. The company's sales manager believes that an increase of \$10,000 in the monthly advertising budget would increase bike sales to 540 units.

- Should we authorize the requested increase in the advertising budget?

Changes in Fixed Costs and Sales Volume

\$80,000 + \$10,000 advertising = \$90,000

	Current Sales (500 bikes)	Projected Sales (540 bikes)
Sales	\$ 250,000	\$ 270,000
Less: variable expenses	150,000	162,000
Contribution margin	100,000	108,000
Less: fixed expenses	80,000	90,000
Net income	\$ 20,000	\$ 18,000

Sales increased by \$20,000, but net income **decreased by \$2,000.**

Changes in Fixed Costs and Sales Volume

The Shortcut Solution

Increase in CM (40 units X \$200)	\$ 8,000
Increase in advertising expenses	10,000
Decrease in net income	<u>\$ (2,000)</u>

Break-Even Analysis

Break-even analysis can be approached in two ways:

- ① Equation method
- ② Contribution margin method.

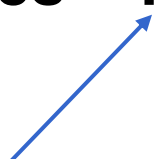
Equation Method

$$\text{Profits} = \text{Sales} - (\text{Variable expenses} + \text{Fixed expenses})$$

OR

$$\text{Sales} = \text{Variable expenses} + \text{Fixed expenses} + \text{Profits}$$

At the break-even point
profits equal zero.



Equation Method

Here is the information from Wind Bicycle Co.:

	<u>Total</u>	<u>Per Unit</u>	<u>Percent</u>
Sales (500 bikes)	\$250,000	\$ 500	100%
Less: variable expenses	150,000	300	60%
Contribution margin	\$100,000	\$ 200	40%
Less: fixed expenses	80,000		
Net income	\$ 20,000		



Equation Method

We calculate the break-even point as follows:

$$\text{Sales} = \text{Variable expenses} + \text{Fixed expenses} + \text{Profits}$$

$$\$500Q = \$300Q + \$80,000 + \$0$$

Where:

Q	= Number of bikes sold
\$500	= Unit sales price
\$300	= Unit variable expenses
\$80,000	= Total fixed expenses



Equation Method

We calculate the break-even point as follows:

$$\text{Sales} = \text{Variable expenses} + \text{Fixed expenses} + \text{Profits}$$

$$\$500Q = \$300Q + \$80,000 + \$0$$

$$\$200Q = \$80,000$$

$$Q = 400 \text{ bikes}$$

1. Contribution Margin (CM)

Contribution Margin (Total)

Contribution Margin = Sales - Variable Expenses

Contribution Margin = Sales - Variable Expenses

Contribution Margin (Per Unit)

CM per unit = Selling Price per Unit - Variable Cost per Unit

CM per unit = Selling Price per Unit - Variable Cost per Unit

2. Net Income (Contribution Format Income Statement)

Net Income = Contribution Margin - Fixed Expenses

Net Income = Contribution Margin - Fixed Expenses

3. Contribution Margin Ratio (CM Ratio)

CM Ratio = Contribution Margin / Sales

CM Ratio =

Selling Price per Unit

Contribution Margin

or (per unit form)

CM Ratio = CM per unit / Selling Price per unit

CM Ratio =

Selling Price per Unit



Equation Method

- We can also use the following equation to compute the break-even point in **sales dollars**.

$$\text{Sales} = \text{Variable expenses} + \text{Fixed expenses} + \text{Profits}$$

$$X = 0.60X + \$80,000 + \$0$$

Where:

X = Total **sales dollars**

0.60 = Variable expenses as a percentage of sales

$\$80,000$ = Total fixed expenses



Equation Method

- We can also use the following equation to compute the break-even point in **sales dollars**.

Sales = Variable expenses + Fixed expenses + Profits

$$X = 0.60X + \$80,000 + \$0$$

$$0.40X = \$80,000$$

$$X = \$200,000$$



Contribution Margin Method

The contribution margin method is a variation of the equation method.

$$\text{Break-even point in units sold} = \frac{\text{Fixed expenses}}{\text{Unit contribution margin}}$$

$$\text{Break-even point in total sales dollars} = \frac{\text{Fixed expenses}}{\text{CM ratio}}$$

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. 2,100 cups are sold each month on average. What is the break-even sales in units?

- a. 872 cups
- b. 3,611 cups
- c. 1,200 cups
- d. 1,150 cups

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. 2,100 cups are sold each month on average. What is the break-even sales in dollars?

- a. \$1,300
- b. \$1,715
- c. \$1,788
- d. \$3,129

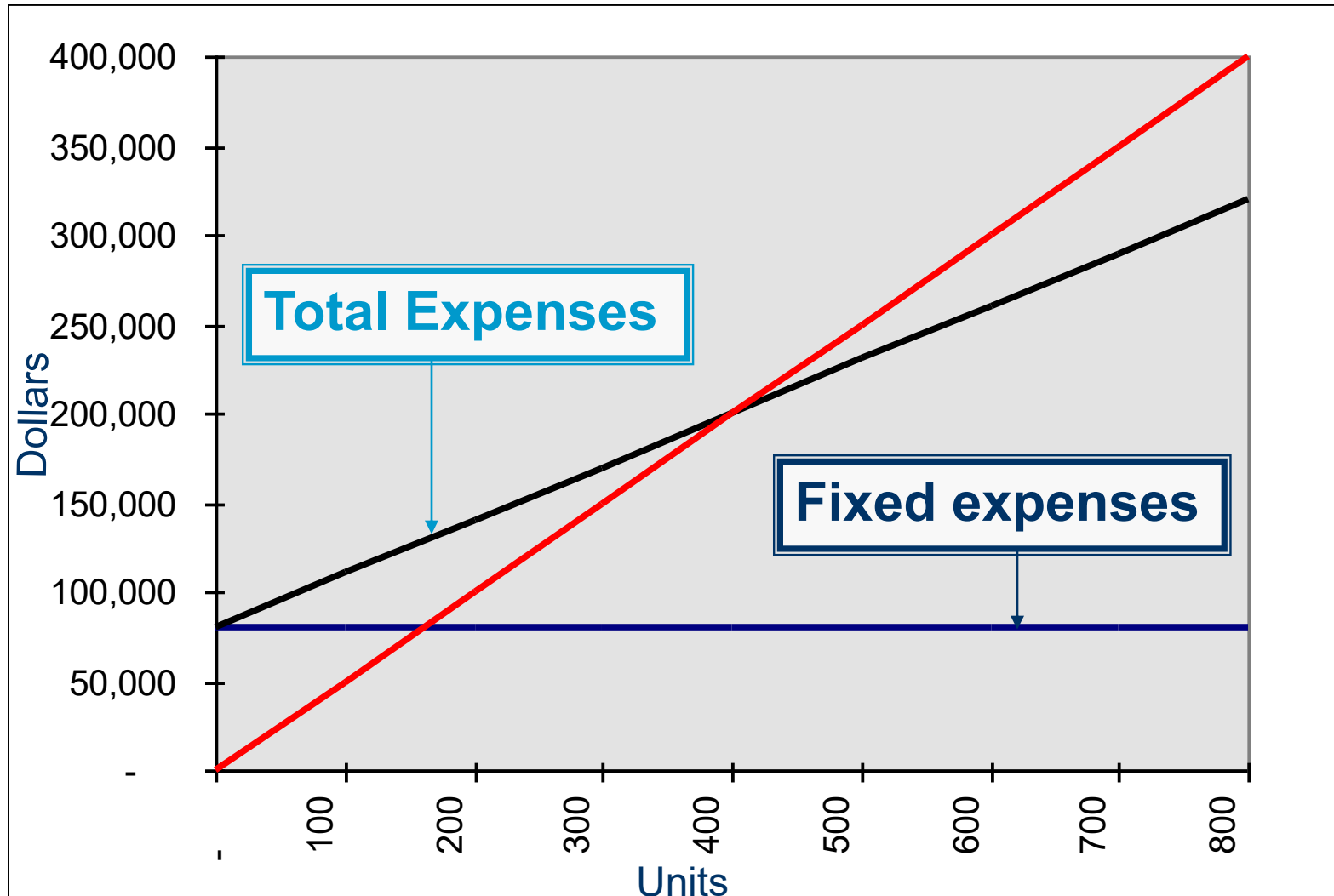
CVP Relationships in Graphic Form

Viewing CVP relationships in a graph gives managers a perspective that can be obtained in no other way.
Consider the following information for Wind Co.:

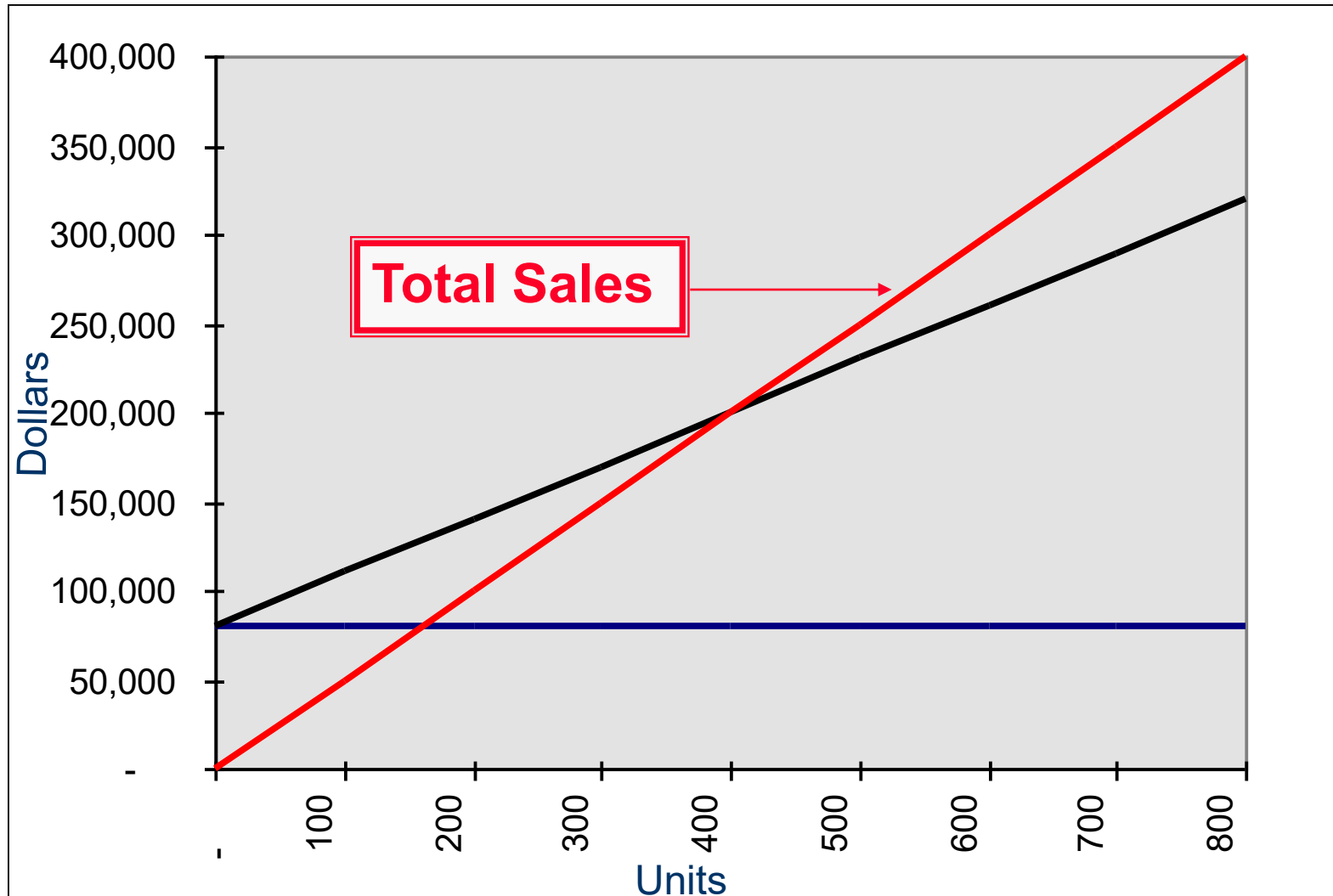
	Income 300 units	Income 400 units	Income 500 units
Sales	\$ 150,000	\$ 200,000	\$250,000
Less: variable expenses	90,000	120,000	150,000
Contribution margin	\$ 60,000	\$ 80,000	\$100,000
Less: fixed expenses	80,000	80,000	80,000
Net income (loss)	\$ (20,000)	\$ -	\$ 20,000



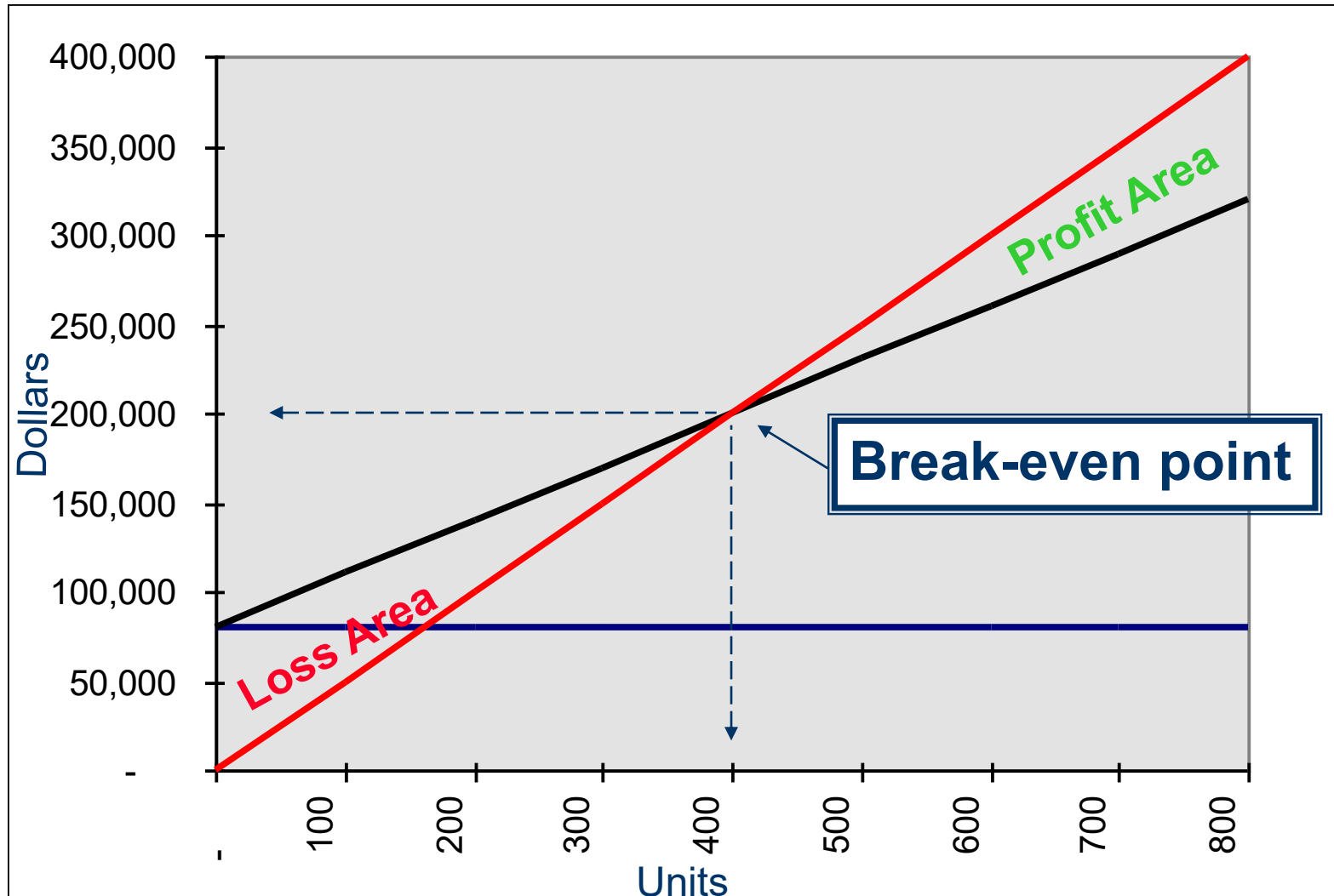
CVP Graph



CVP Graph



CVP Graph



Target Profit Analysis

Suppose Wind Co. wants to know how many bikes must be sold to earn a profit of \$100,000.

We can use our CVP formula to determine the sales volume needed to achieve a target net profit figure.



The CVP Equation

Sales = Variable expenses + Fixed expenses + Profits

$$\text{\$500Q} = \text{\$300Q} + \text{\$80,000} + \text{\$100,000}$$

$$\text{\$200Q} = \text{\$180,000}$$

$$Q = 900 \text{ bikes}$$



The Contribution Margin Approach

We can determine the number of bikes that must be sold to earn a profit of \$100,000 using the contribution margin approach.

$$\text{Units sold to attain the target profit} = \frac{\text{Fixed expenses} + \text{Target profit}}{\text{Unit contribution margin}}$$

$$\frac{\$80,000 + \$100,000}{\$200} = 900 \text{ bikes}$$



Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. How many cups of coffee would have to be sold to attain target profits of \$2,500 per month?

- a. 3,363 cups
- b. 2,212 cups
- c. 1,150 cups
- d. 4,200 cups



The Margin of Safety

Excess of budgeted (or actual) sales over the break-even volume of sales. The amount by which sales can drop before losses begin to be incurred.

Margin of safety = Total sales - Break-even sales

Let's calculate the margin of safety for Wind.

The Margin of Safety

Wind has a break-even point of \$200,000. If actual sales are \$250,000, the margin of safety is **\$50,000** or 100 bikes.

	Break-even sales 400 units	Actual sales 500 units
Sales	\$ 200,000	\$ 250,000
Less: variable expenses	120,000	150,000
Contribution margin	80,000	100,000
Less: fixed expenses	80,000	80,000
Net income	\$ -	\$ 20,000



The Margin of Safety

The margin of safety can be expressed as
20 percent of sales.
($\$50,000 \div \$250,000$)

	Break-even sales 400 units	Actual sales 500 units
Sales	\$ 200,000	\$ 250,000
Less: variable expenses	120,000	150,000
Contribution margin	80,000	100,000
Less: fixed expenses	80,000	80,000
Net income	\$ -	\$ 20,000



Quick Check ✓

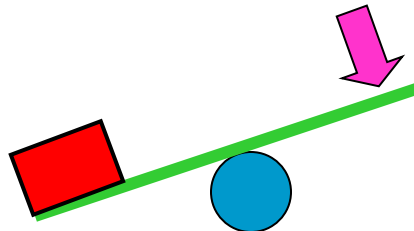
Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. 2,100 cups are sold each month on average. What is the margin of safety?

- a. 3,250 cups
- b. 950 cups
- c. 1,150 cups
- d. 2,100 cups

Operating Leverage

- A measure of how sensitive net income is to percentage changes in sales.
- With high leverage, a small percentage increase in sales can produce a much larger percentage increase in net income.

$$\text{Degree of operating leverage} = \frac{\text{Contribution margin}}{\text{Net income}}$$



Operating Leverage

	Actual sales 500 Bikes
Sales	\$ 250,000
Less: variable expenses	150,000
Contribution margin	100,000
Less: fixed expenses	80,000
Net income	\$ 20,000

$$\frac{\$100,000}{\$20,000} = 5$$



Operating Leverage

With a measure of operating leverage of 5, if Wind increases its sales by 10%, net income would increase by 50%.

Percent increase in sales		10%
Degree of operating leverage	×	<u>5</u>
Percent increase in profits		<u><u>50%</u></u>

Here's the proof!



Operating Leverage

	Actual sales (500)	Increased sales (550)
Sales	\$ 250,000	\$ 275,000
Less variable expenses	150,000	165,000
Contribution margin	100,000	110,000
Less fixed expenses	80,000	80,000
Net income	\$ 20,000	\$ 30,000

**10% increase in sales from
\$250,000 to \$275,000 . . .**

**. . . results in a 50% increase in
income from \$20,000 to \$30,000.**

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. 2,100 cups are sold each month on average. What is the operating leverage?

- a. 2.21
- b. 0.45
- c. 0.34
- d. 2.92

Quick Check ✓

At Coffee Klatch the average selling price of a cup of coffee is \$1.49, the average variable expense per cup is \$0.36, and the average fixed expense per month is \$1,300. 2,100 cups are sold each month on average.

If sales increase by 20%, by how much should net income increase?

- a. 30.0%
- b. 20.0%
- c. 22.1%
- d. 44.2%

The Concept of Sales Mix

- Sales mix is the relative proportions in which a company's products are sold.
- Different products have different selling prices, cost structures, and contribution margins.

Let's assume Wind sells bikes and carts and see how we deal with break-even analysis.



Multi-product break-even analysis

Wind Bicycle Co. provides the following information:

	Bikes			Carts			Total	
Sales	\$	250,000	100%	\$	300,000	100%	\$	550,000 100.0%
Var. exp.		150,000	60%		135,000	45%		285,000 51.8%
Contrib. margin	\$	100,000	40%	\$	165,000	55%		265,000 48.2%
Fixed exp.								170,000
Net income								\$ 95,000
Sales mix	\$	250,000	45%	\$	300,000	55%	\$	550,000 100.0%

$$\frac{\$265,000}{\$550,00} = 48.2\% \text{ (rounded)}$$



Multi-product break-even analysis

$$\begin{aligned}\text{Breakeven sales} &= \frac{F}{\text{CM Ratio}} \\ &= \frac{\$170,000}{0.482} = \$352,697\end{aligned}$$

	Bikes		Carts		Total	
Sales	\$ 158,714	100%	\$ 193,983	100%	\$ 352,697	100.0%
Var. exp.	95,228	60%	87,293	45%	182,521	51.8%
Contrib. margin	<u>\$ 63,485</u>	<u>40%</u>	<u>\$ 106,691</u>	<u>55%</u>	<u>170,176</u>	<u>48.2%</u>
Fixed exp.					<u>170,000</u>	
Net income					<u>\$ 176</u>	
Rounding error						
Sales mix	\$ 158,714	45%	\$ 193,983	55%	\$ 352,697	100.0%



Assumptions of CVP Analysis

- 1 Selling price is constant throughout the entire relevant range.
- 2 Costs are linear throughout the entire relevant range.
- 3 In multi-product companies, the sales mix is constant.
- 4 In manufacturing companies, inventories do not change (units produced = units sold).

